DIGITAL USAGE INTENSITY INDEX(DUII)

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Tools Used-

- Jupyter Notebook(For Python)
- Numpy, Pandas (Feature engineering, Dataset)
- Seaborn (Plotting)
- Matplotlib (Plotting)
- Sklearn (For scaler)

STEP1:

- Setup(Jupyter Notebook is used for this Data Science Project for visualization,feature engineering etc..)
- Loading Data(Data here is obtained from kaggle.com)

[1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

[2]:
```
df1=pd.read_csv('digital_diet_mental_health.csv')
```

[3]:
```
df1.head()
```

[3]:

	user_id	age	gender	daily_screen_time_hours	phone_usage_hours	laptop_usage_hours	tablet_usage_hours	tv_usage_hours	social_media_hours	work_related_hours
0	user_1	51	Female	4.8	3.4	1.3	1.6	1.6	4.1	2.0
1	user_2	64	Male	3.9	3.5	1.8	0.9	2.0	2.7	3.1
2	user_3	41	Other	10.5	2.1	2.6	0.7	2.2	3.0	2.8
3	user_4	27	Other	8.8	0.0	0.0	0.7	2.5	3.3	1.6
4	user_5	55	Male	5.9	1.7	1.1	1.5	1.6	1.1	3.6

5 rows × 25 columns

[4]:
```
df1.columns
```

[4]:
```
Index(['user_id', 'age', 'gender', 'daily_screen_time_hours',
       'phone_usage_hours', 'laptop_usage_hours', 'tablet_usage_hours',
       'tv_usage_hours', 'social_media_hours', 'work_related_hours',
       'entertainment_hours', 'gaming_hours', 'sleep_duration_hours',
       'sleep_quality', 'mood_rating', 'stress_level',
       'physical_activity_hours_per_week', 'location_type',
       'mental_health_score', 'uses_wellness_apps', 'eats_healthy',
       'caffeine_intake_mg_per_day', 'weekly_anxiety_score',
       'weekly_depression_score', 'mindfulness_minutes_per_day'],
      dtype='object')
```

[5]:
```
df2=pd.read_csv('screentime_analysis.csv')
```

[6]:
```
df2.head()
```

[6]:

	Date	App	Usage (minutes)	Notifications	Times Opened
0	2024-08-07	Instagram	81	24	57
1	2024-08-08	Instagram	90	30	53
2	2024-08-26	Instagram	112	33	17
3	2024-08-22	Instagram	82	11	38
4	2024-08-12	Instagram	59	47	16

[7]:
```
df2.columns
```

[7]:
```
Index(['Date', 'App', 'Usage (minutes)', 'Notifications', 'Times Opened'], dtype='object')
```

[8]:
```
df3=pd.read_csv('Students Social Media Addiction.csv')
```

[9]:
```
df3.head()
```

[9]:

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performance	Sleep_Hours_Per_Night	Mental
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	Yes	6.5	
1	2	22	Male	Graduate	India	2.1	Twitter	No	7.5	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	Yes	5.0	
3	4	18	Male	High School	UK	3.0	YouTube	No	7.0	
4	5	21	Male	Graduate	Canada	4.5	Facebook	Yes	6.0	

[10]:
```
df3.columns
```

[10]:
```
Index(['Student_ID', 'Age', 'Gender', 'Academic_Level', 'Country',
       'Avg_Daily_Usage_Hours', 'Most_Used_Platform',
       'Affects_Academic_Performance', 'Sleep_Hours_Per_Night',
       'Mental_Health_Score', 'Relationship_Status',
       'Conflicts_Over_Social_Media', 'Addicted_Score'],
      dtype='object')
```

```
[11]: df4=pd.read_csv('teen_phone_addiction_dataset.csv')
```

```
[12]: df4.head()
```

[12]:

	ID	Name	Age	Gender	Location	School_Grade	Daily_Usage_Hours	Sleep_Hours	Academic_Performance	Social_Interactions	...	Screen_Time_Before_Bed
0	1	Shannon Francis	13	Female	Hansonfort	9th	4.0	6.1	78	5	...	1.4
1	2	Scott Rodriguez	17	Female	Theodorefort	7th	5.5	6.5	70	5	...	0.9
2	3	Adrian Knox	13	Other	Lindseystad	11th	5.8	5.5	93	8	...	0.5
3	4	Brittany Hamilton	18	Female	West Anthony	12th	3.1	3.9	78	8	...	1.4
4	5	Steven Smith	14	Other	Port Lindsaystad	9th	2.5	6.7	56	4	...	1.0

5 rows × 25 columns

```
[13]: df4.columns
```

```
[13]: Index(['ID', 'Name', 'Age', 'Gender', 'Location', 'School_Grade',
       'Daily_Usage_Hours', 'Sleep_Hours', 'Academic_Performance',
       'Social_Interactions', 'Exercise_Hours', 'Anxiety_Level',
       'Depression_Level', 'Self_Esteem', 'Parental_Control',
       'Screen_Time_Before_Bed', 'Phone_Checks_Per_Day', 'Apps_Used_Daily',
       'Time_on_Social_Media', 'Time_on_Gaming', 'Time_on_Education',
       'Phone_Usage_Purpose', 'Family_Communication', 'Weekend_Usage_Hours',
       'Addiction_Level'],
      dtype='object')
```

STEP2:

- Remove Duplicate data and null values

  <u>DATASET 1</u>

```
[18]: df1.isnull()
```

[18]:

	user_id	age	gender	daily_screen_time_hours	phone_usage_hours	laptop_usage_hours	tablet_usage_hours	tv_usage_hours	social_media_hours	work_related_h
0	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	...	
1995	False	False	False	False	False	False	False	False	False	
1996	False	False	False	False	False	False	False	False	False	
1997	False	False	False	False	False	False	False	False	False	
1998	False	False	False	False	False	False	False	False	False	
1999	False	False	False	False	False	False	False	False	False	

2000 rows × 25 columns

```
[24]: df1.isnull().sum()
```

```
[24]: user_id                          0
      age                              0
      gender                           0
      daily_screen_time_hours          0
      phone_usage_hours                0
      laptop_usage_hours               0
      tablet_usage_hours               0
      tv_usage_hours                   0
      social_media_hours               0
      work_related_hours               0
      entertainment_hours              0
      gaming_hours                     0
      sleep_duration_hours             0
      sleep_quality                    0
      mood_rating                      0
      stress_level                     0
      physical_activity_hours_per_week 0
      location_type                    0
      mental_health_score              0
      uses_wellness_apps               0
      eats_healthy                     0
      caffeine_intake_mg_per_day       0
      weekly_anxiety_score             0
      weekly_depression_score          0
      mindfulness_minutes_per_day      0
      dtype: int64
```

→This shows us that there no null values present in the dataset1

```
[25]: df1.duplicated().sum()

[25]: 0
```

→This Shows there are no duplicate rows in the dataset1

DATASET 2

```
[26]: df2.isnull()
```

	Date	App	Usage (minutes)	Notifications	Times Opened
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
195	False	False	False	False	False
196	False	False	False	False	False
197	False	False	False	False	False
198	False	False	False	False	False
199	False	False	False	False	False

200 rows × 5 columns

```
[28]: df2.isnull().sum()

[28]: Date              0
      App               0
      Usage (minutes)   0
      Notifications     0
      Times Opened      0
      dtype: int64

[29]: df2.duplicated().sum()

[29]: 0
```

→This shows that there are no null and duplicate values in dataset2

DATASET 3

```
[30]: df3.isnull()
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performance	Sleep_Hours_Per_Night	Menta
0	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	...	
700	False	False	False	False	False	False	False	False	False	
701	False	False	False	False	False	False	False	False	False	
702	False	False	False	False	False	False	False	False	False	
703	False	False	False	False	False	False	False	False	False	
704	False	False	False	False	False	False	False	False	False	

705 rows × 13 columns

```
[31]: df3.isnull().sum()

[31]: Student_ID                    0
      Age                           0
      Gender                        0
      Academic_Level                0
      Country                       0
      Avg_Daily_Usage_Hours         0
      Most_Used_Platform            0
      Affects_Academic_Performance  0
      Sleep_Hours_Per_Night         0
      Mental_Health_Score           0
      Relationship_Status           0
      Conflicts_Over_Social_Media   0
      Addicted_Score                0
      dtype: int64

[32]: df3.duplicated().sum()

[32]: 0
```

→This shows that there are no null and duplicate values in dataset3

DATASET 4

```
[34]: df4.isnull()
```

[34]:

	ID	Name	Age	Gender	Location	School_Grade	Daily_Usage_Hours	Sleep_Hours	Academic_Performance	Social_Interactions	...	Screen_Time_Before_Bed
0	False	False	False	False	False	False	False	False	False	False	...	False
1	False	False	False	False	False	False	False	False	False	False	...	False
2	False	False	False	False	False	False	False	False	False	False	...	False
3	False	False	False	False	False	False	False	False	False	False	...	False
4	False	False	False	False	False	False	False	False	False	False	...	False
...	...	...	...	...	...	...	...	...	...	...	...	...
2995	False	False	False	False	False	False	False	False	False	False	...	False
2996	False	False	False	False	False	False	False	False	False	False	...	False
2997	False	False	False	False	False	False	False	False	False	False	...	False
2998	False	False	False	False	False	False	False	False	False	False	...	False
2999	False	False	False	False	False	False	False	False	False	False	...	False

3000 rows × 25 columns

```
[35]: df4.isnull().sum()
```

```
[35]: ID                       0
      Name                     0
      Age                      0
      Gender                   0
      Location                 0
      School_Grade             0
      Daily_Usage_Hours        0
      Sleep_Hours              0
      Academic_Performance     0
      Social_Interactions      0
      Exercise_Hours           0
      Anxiety_Level            0
      Depression_Level         0
      Self_Esteem              0
      Parental_Control         0
      Screen_Time_Before_Bed   0
      Phone_Checks_Per_Day     0
      Apps_Used_Daily          0
      Time_on_Social_Media     0
      Time_on_Gaming           0
      Time_on_Education        0
      Phone_Usage_Purpose      0
      Family_Communication     0
      Weekend_Usage_Hours      0
      Addiction_Level          0
      dtype: int64
```

```
[36]: df4.duplicated().sum()
```

```
[36]: 0
```

→This shows that there are no null and duplicate values in dataset4

STEP3:

- Feature Selection and Variable Categorization i.e grouping columns into validation and core columns
  -Core columns which will be used for the analysis and then to compute Digital Usage Intensity Index (DUII). They represent measurable behavioural dimensions

-<u>Validation columns</u> which will be used to validate and check if the obtained Digital Usage Intensity Index (DUII) holds good. These variables are retained for post-construction validation

Important Principle in Metric Design

Only include variables that:

✓ Directly measure the construct

✓ Represent the phenomenon

✓ Are conceptually aligned

<u>DATASET 1</u>

```
[37]: core_cols_df1 = ['daily_screen_time_hours','social_media_hours','sleep_duration_hours','stress_level']

      validation_cols_df1 = ['mood_rating','mental_health_score']

      df1_core = df1[core_cols_df1].copy()
      df1_validation = df1[validation_cols_df1].copy()
```

```
[38]: df1_core.head()
```

[38]:

	daily_screen_time_hours	social_media_hours	sleep_duration_hours	stress_level
0	4.8	4.1	6.6	10
1	3.9	2.7	4.5	6
2	10.5	3.0	7.1	5
3	8.8	3.3	5.1	5
4	5.9	1.1	7.4	7

```
[40]: df1_validation.head()
```

[40]:

	mood_rating	mental_health_score
0	6	32
1	5	75
2	5	22
3	10	22
4	8	64

→Here some columns like for example age is a demographic value and not a behavioural usage variable so it does not represent Usage duration, Dependency or Disruption.

<u>DATASET 2</u>

```
df2_grouped = df2.groupby('App').agg({
    'Usage (minutes)': 'sum',
    'Notifications': 'sum',
    'Times Opened': 'sum'
}).reset_index()
```

```
df2_grouped.head()
```

	App	Usage (minutes)	Notifications	Times Opened
0	8 Ball Pool	452	113	182
1	Facebook	842	993	755
2	Instagram	1898	1245	1039
3	LinkedIn	390	223	119
4	Netflix	1819	11	64

DATASET 3

```
[45]: df3['Affects_Academic_Performance'] = df3['Affects_Academic_Performance'].map({'Yes':1,'No':0})
[46]: df3.head()
```

[46]:

Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performance	Sleep_Hours_Per_Night	Mental_Health_Score	Relationship_Status	Conflicts_Over_Social_M
5.2	Instagram	1	6.5	6	In Relationship	
2.1	Twitter	0	7.5	8	Single	
6.0	TikTok	1	5.0	5	Complicated	
3.0	YouTube	0	7.0	7	Single	
4.5	Facebook	1	6.0	6	In Relationship	

```
[ ]:
```

->This Changes the values inside the Affects_Academic_Performance from YES/NO to 1/0 for better visualization and index calculation (Text Vectorization)

```
df3_core = df3[['Avg_Daily_Usage_Hours','Sleep_Hours_Per_Night','Addicted_Score']].copy()
```

```
df3_core.head()
```

	Avg_Daily_Usage_Hours	Sleep_Hours_Per_Night	Addicted_Score
0	5.2	6.5	8
1	2.1	7.5	3
2	6.0	5.0	9
3	3.0	7.0	4
4	4.5	6.0	7

->Core dataset3 is made we have excluded some columns which not will be any useful like platform used and relationsihp status does not affect our formula in a large scale.

DATASET 4

```
[49]: df4_core = df4[['Daily_Usage_Hours','Phone_Checks_Per_Day','Screen_Time_Before_Bed','Addiction_Level']].copy()
```

```
[50]: df4_core.head()
```

[50]:

	Daily_Usage_Hours	Phone_Checks_Per_Day	Screen_Time_Before_Bed	Addiction_Level
0	4.0	86	1.4	10.0
1	5.5	96	0.9	10.0
2	5.8	137	0.5	9.2
3	3.1	128	1.4	9.8
4	2.5	96	1.0	8.6

->Core dataset4 is made we have excluded some columns which will not be any useful like depression level, anxiety level, self-esteem or parental control doesn't etc will not affect the index much.

STEP4:

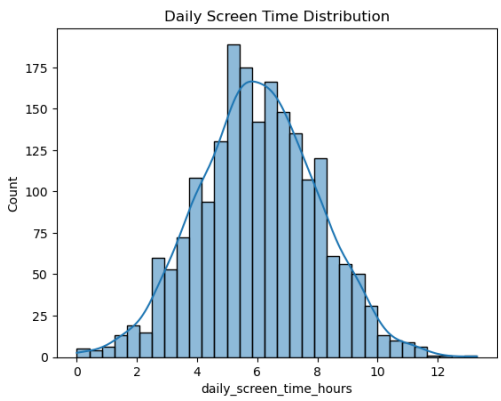
- Basic EDA (Exploratory Data Analysis)(Raw Visualization)

  <u>DATASET 1</u>

  Distribution Plots

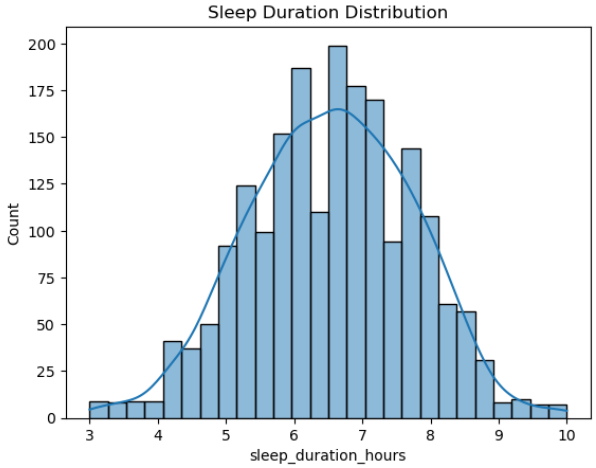
```
[52]: sns.histplot(df1['daily_screen_time_hours'], kde=True)
      plt.title("Daily Screen Time Distribution")
```

```
[52]: Text(0.5, 1.0, 'Daily Screen Time Distribution')
```



```
[53]: sns.histplot(df1['sleep_duration_hours'], kde=True)
      plt.title("Sleep Duration Distribution")
```

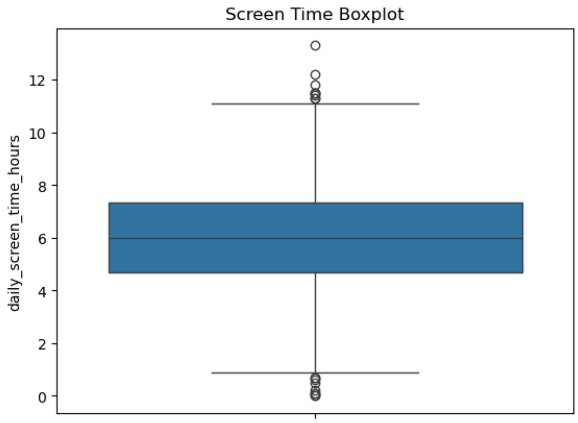
```
[53]: Text(0.5, 1.0, 'Sleep Duration Distribution')
```



Boxplots (Outlier Check)

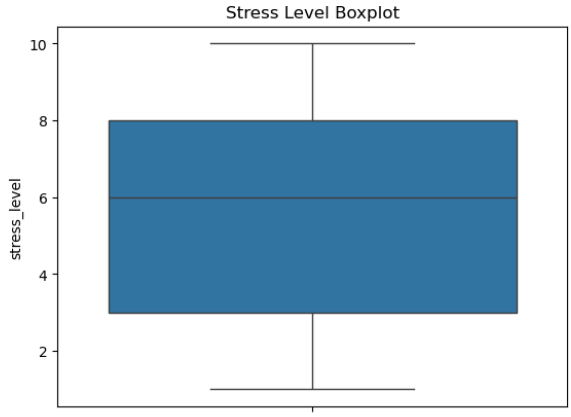
```
[54]: sns.boxplot(y=df1['daily_screen_time_hours'])
      plt.title("Screen Time Boxplot")
```

```
[54]: Text(0.5, 1.0, 'Screen Time Boxplot')
```




```
[55]: sns.boxplot(y=df1['stress_level'])
      plt.title("Stress Level Boxplot")
```

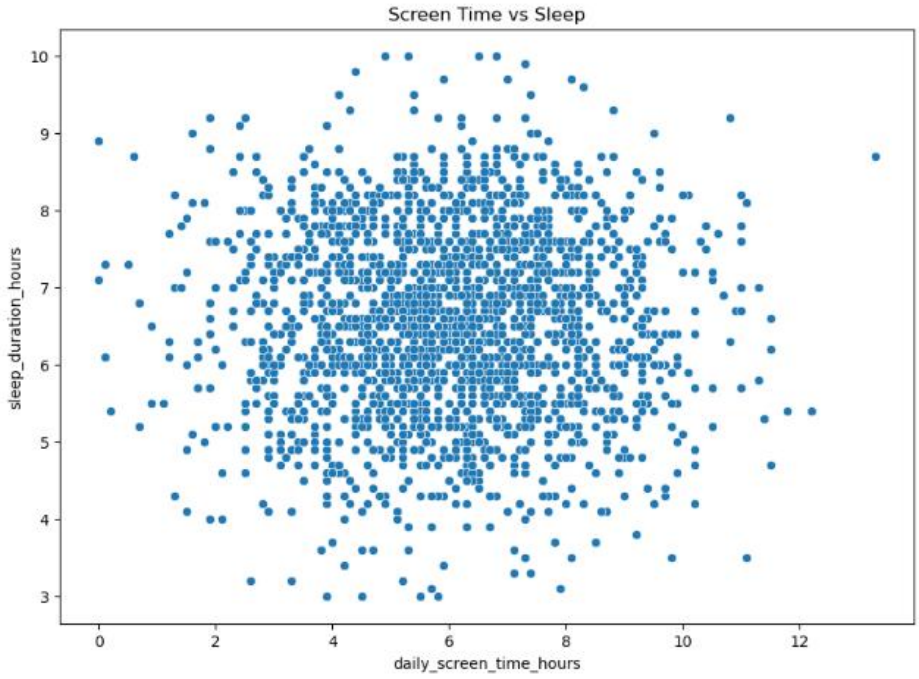
```
[55]: Text(0.5, 1.0, 'Stress Level Boxplot')
```



Relationship Plot

```
[58]: plt.figure(figsize=(10,7))
      plt.tight_layout()
      sns.scatterplot(x='daily_screen_time_hours',y='sleep_duration_hours',data=df1)
      plt.title("Screen Time vs Sleep")
```

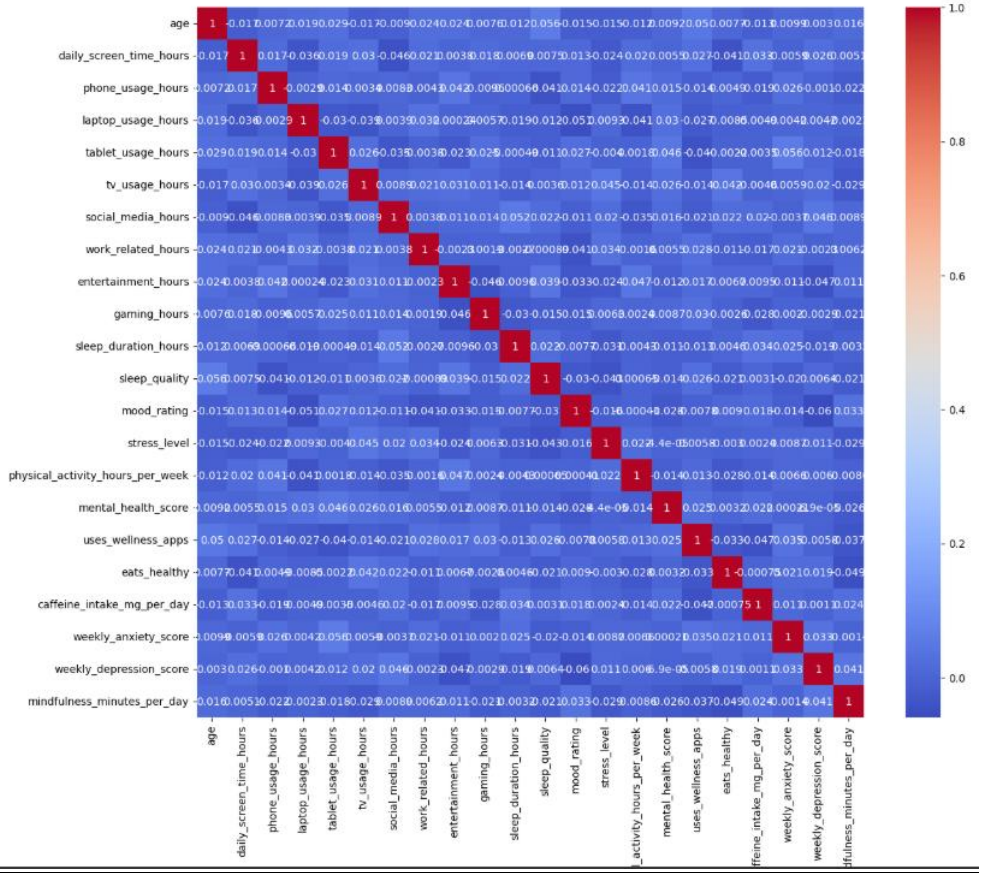
```
[58]: Text(0.5, 1.0, 'Screen Time vs Sleep')
```



Correlation Heatmap

```
[61]: plt.figure(figsize=(14,12))
      plt.tight_layout()
      sns.heatmap(df1.corr(numeric_only=True),annot=True,cmap="coolwarm")
```

[61]: <Axes: >



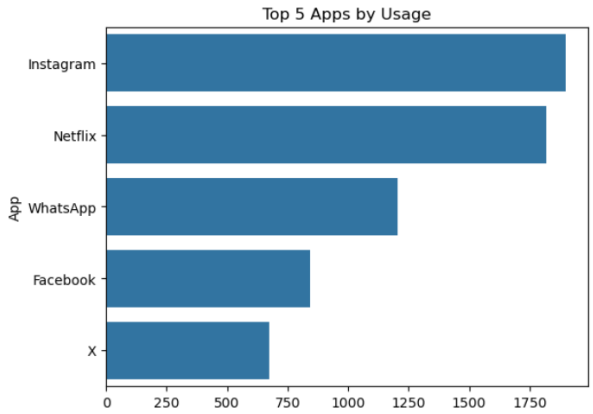
DATASET 2

Top Apps by Usage

[62]:
```
app_usage = df2.groupby('App')['Usage (minutes)'].sum().sort_values(ascending=False)
```

[63]:
```
sns.barplot(x=app_usage.values[:5],y=app_usage.index[:5])
plt.title("Top 5 Apps by Usage")
```

[63]: Text(0.5, 1.0, 'Top 5 Apps by Usage')

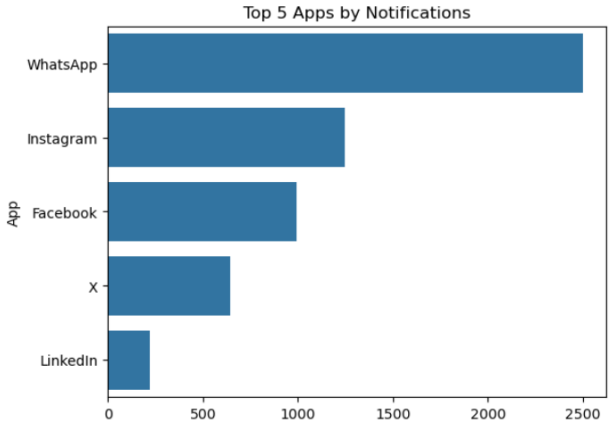


Notifications per App

```
[64]: notif = df2.groupby('App')['Notifications'].sum().sort_values(ascending=False)
```

```
[65]: sns.barplot(x=notif.values[:5],y=notif.index[:5])
      plt.title("Top 5 Apps by Notifications")
```

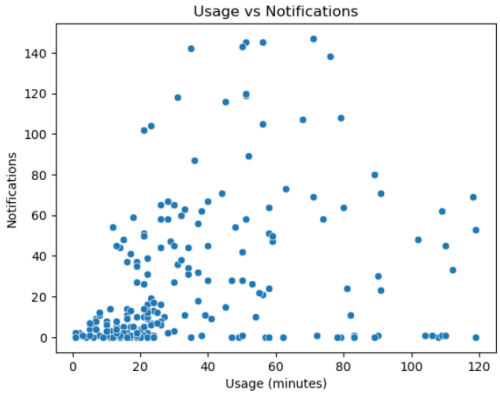
```
[65]: Text(0.5, 1.0, 'Top 5 Apps by Notifications')
```



Engagement Scatter

```
[66]: sns.scatterplot(x='Usage (minutes)',y='Notifications',data=df2)
      plt.title("Usage vs Notifications")
```

```
[66]: Text(0.5, 1.0, 'Usage vs Notifications')
```

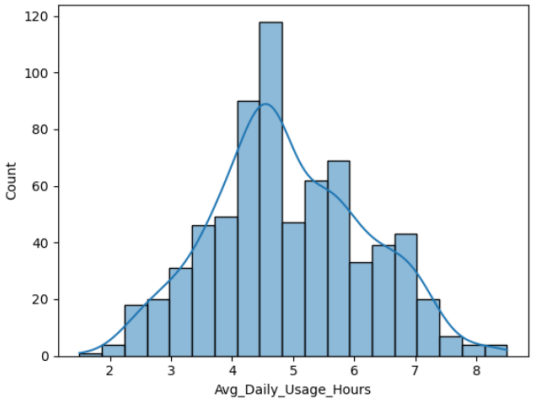


DATASET 3

Usage Distribution

```
[67]: sns.histplot(df3['Avg_Daily_Usage_Hours'], kde=True)
```

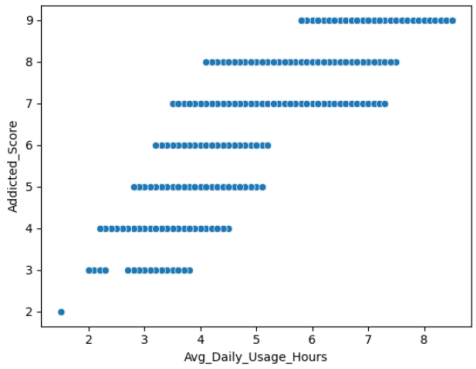
```
[67]: <Axes: xlabel='Avg_Daily_Usage_Hours', ylabel='Count'>
```



Usage vs Addiction

```
[68]: sns.scatterplot(x='Avg_Daily_Usage_Hours',y='Addicted_Score',data=df3)
```

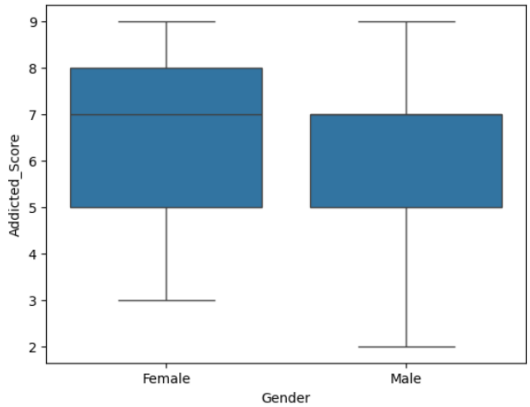
```
[68]: <Axes: xlabel='Avg_Daily_Usage_Hours', ylabel='Addicted_Score'>
```



Addiction by Gender

```
[69]: sns.boxplot(x='Gender',y='Addicted_Score',data=df3)
```

```
[69]: <Axes: xlabel='Gender', ylabel='Addicted_Score'>
```

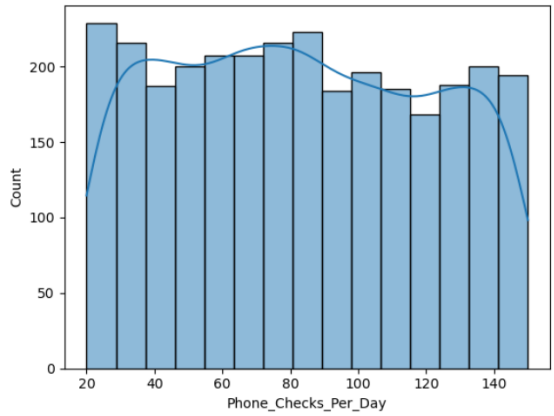


DATASET 4

Phone Checks Distribution

```
[70]: sns.histplot(df4['Phone_Checks_Per_Day'], kde=True)
```

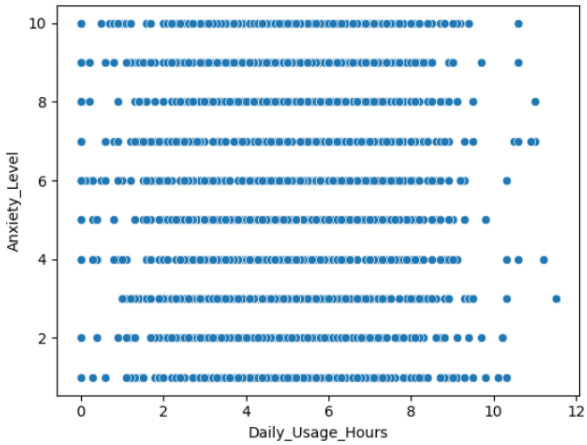
```
[70]: <Axes: xlabel='Phone_Checks_Per_Day', ylabel='Count'>
```



Usage vs Anxiety

```
[71]: sns.scatterplot(x='Daily_Usage_Hours',y='Anxiety_Level',data=df4)
```

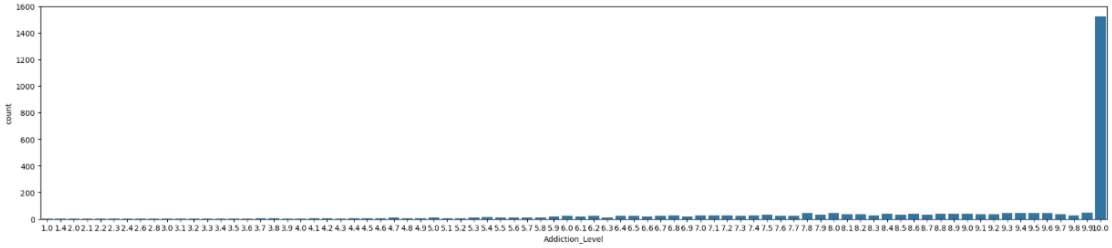
```
[71]: <Axes: xlabel='Daily_Usage_Hours', ylabel='Anxiety_Level'>
```



Addiction Level Count

```
[77]: plt.figure(figsize=(25,5))
plt.tight_layout()
sns.countplot(x='Addiction_Level', data=df4)
```

```
[77]: <Axes: xlabel='Addiction_Level', ylabel='count'>
```



STEP5:

- Feature Engineering
  New columns are derived from existing columns

DATASET 1

```
[80]: df1_core['social_media_ratio'] = (df1_core['social_media_hours'] /df1_core['daily_screen_time_hours'])
```

```
[81]: df1_core.head()
```

[81]:

	daily_screen_time_hours	social_media_hours	sleep_duration_hours	stress_level	social_media_ratio
0	4.8	4.1	6.6	10	0.854167
1	3.9	2.7	4.5	6	0.692308
2	10.5	3.0	7.1	5	0.285714
3	8.8	3.3	5.1	5	0.375000
4	5.9	1.1	7.4	7	0.186441

---

social_media_ratio = ____social_media_hours____

daily_screen_time_hours

```
[82]: df1_core['sleep_deficit'] = 8 - df1_core['sleep_duration_hours']
      df1_core['sleep_deficit'] = df1_core['sleep_deficit'].clip(lower=0)
```

```
[83]: df1_core.head()
```

[83]:

	daily_screen_time_hours	social_media_hours	sleep_duration_hours	stress_level	social_media_ratio	sleep_deficit
0	4.8	4.1	6.6	10	0.854167	1.4
1	3.9	2.7	4.5	6	0.692308	3.5
2	10.5	3.0	7.1	5	0.285714	0.9
3	8.8	3.3	5.1	5	0.375000	2.9
4	5.9	1.1	7.4	7	0.186441	0.6

→Here 8-sleep_duration_hours is taken as on an average a human being should get a sleep of about 8 hours

DATASET 2

```
]: total_usage = df2_grouped['Usage (minutes)'].sum()
   df2_grouped['usage_ratio'] = df2_grouped['Usage (minutes)'] / total_usage
```

```
]: df2_grouped.head()
```

]:

	App	Usage (minutes)	Notifications	Times Opened	usage_ratio
0	8 Ball Pool	452	113	182	0.059868
1	Facebook	842	993	755	0.111523
2	Instagram	1898	1245	1039	0.251391
3	LinkedIn	390	223	119	0.051656
4	Netflix	1819	11	64	0.240927

→Here some of the columns are aggregated i.e sums are found and also an important usage ratio is calculated that uses the formula.

$$\text{usage_ratio} = \frac{\text{usage(minutes)}}{\text{total_usage}}$$

Where total_usage= sum of all usage in the column

STEP6:

- Standardize & Combine Datasets

DATASET 3

```
[84]: df3_core.rename(columns={
    'Avg_Daily_Usage_Hours': 'daily_usage',
    'Sleep_Hours_Per_Night': 'sleep_hours',
    'Addicted_Score': 'addiction_score'
}, inplace=True)
```

```
[85]: df3_core.head()
```

[85]:

	daily_usage	sleep_hours	addiction_score
0	5.2	6.5	8
1	2.1	7.5	3
2	6.0	5.0	9
3	3.0	7.0	4
4	4.5	6.0	7

DATASET 4

```
[86]: df4_core.rename(columns={
    'Daily_Usage_Hours': 'daily_usage',
    'Phone_Checks_Per_Day': 'phone_checks',
    'Addiction_Level': 'addiction_score'
}, inplace=True)
```

```
[87]: df4_core.head()
```

[87]:

	daily_usage	phone_checks	Screen_Time_Before_Bed	addiction_score
0	4.0	86	1.4	10.0
1	5.5	96	0.9	10.0
2	5.8	137	0.5	9.2
3	3.1	128	1.4	9.8
4	2.5	96	1.0	8.6

STEP7:

- We will be creating 2 sub-indices one based on df3_core and one on df4_core
  Behavioral Dependency Index (BDI) → from df4
  Routine Disruption Index (RDI) → from df3

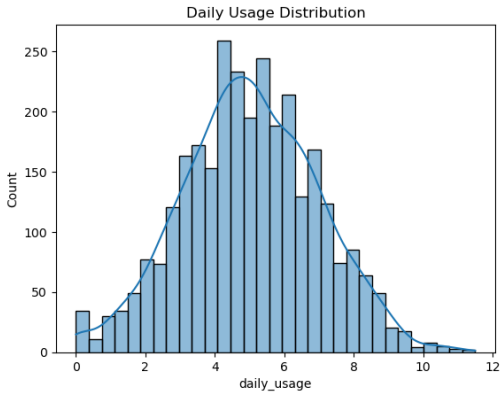
- **ADVANCED EDA ON DF3_CORE,DF4_CORE**

  DATASET4

Distribution Plots

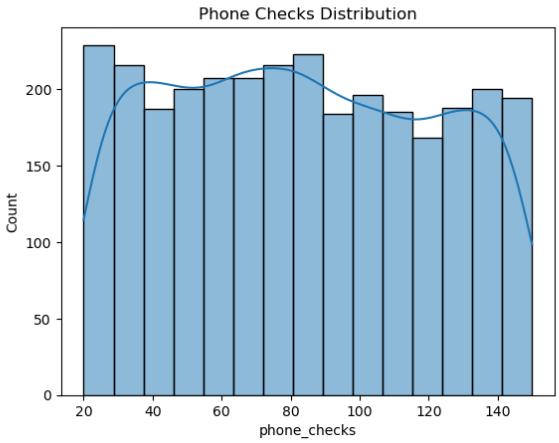
```
[164]: sns.histplot(df4_core['daily_usage'], kde=True)
       plt.title("Daily Usage Distribution")
```

```
[164]: Text(0.5, 1.0, 'Daily Usage Distribution')
```



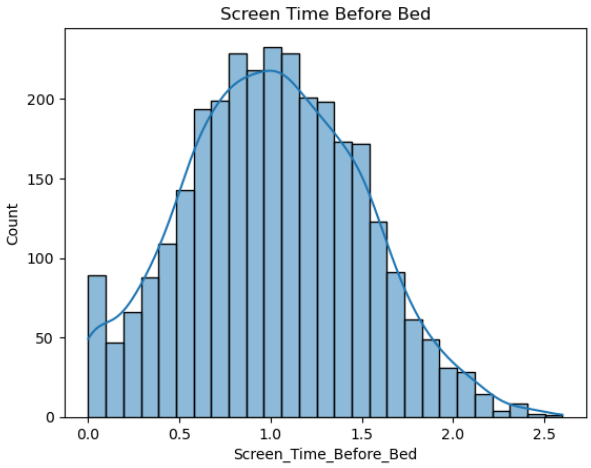

```
[165]: sns.histplot(df4_core['phone_checks'], kde=True)
       plt.title("Phone Checks Distribution")
```

```
[165]: Text(0.5, 1.0, 'Phone Checks Distribution')
```




```
[166]: sns.histplot(df4_core['Screen_Time_Before_Bed'], kde=True)
       plt.title("Screen Time Before Bed")
```

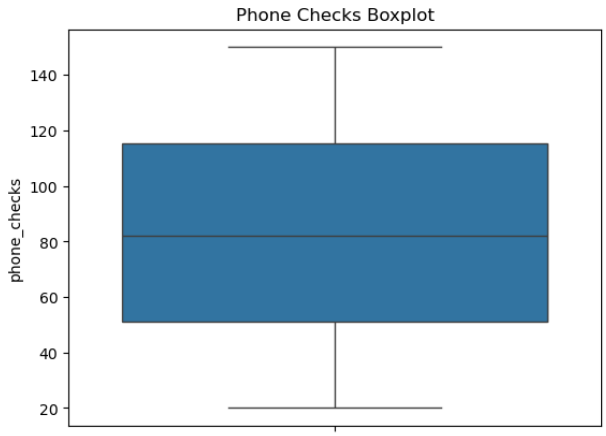
```
[166]: Text(0.5, 1.0, 'Screen Time Before Bed')
```



Outlier Detection

```
[167]: sns.boxplot(y=df4_core['phone_checks'])
       plt.title("Phone Checks Boxplot")
```

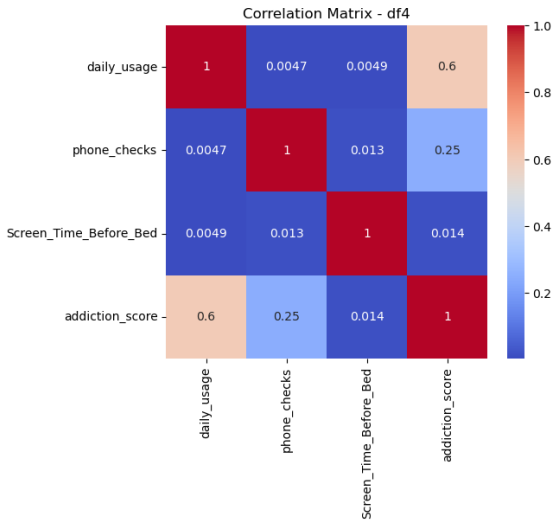
```
[167]: Text(0.5, 1.0, 'Phone Checks Boxplot')
```



Correlation Heatmap

```
[168]: plt.figure(figsize=(6,5))
       sns.heatmap(df4_core.corr(), annot=True, cmap='coolwarm')
       plt.title("Correlation Matrix - df4")
```

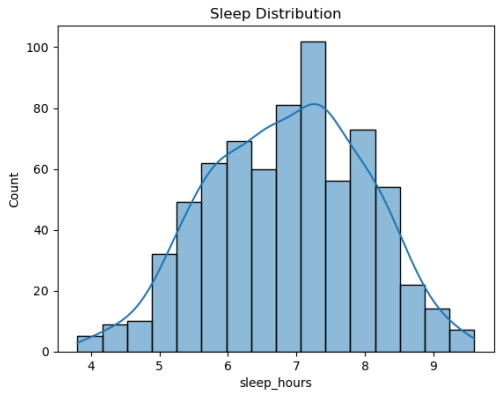
```
[168]: Text(0.5, 1.0, 'Correlation Matrix - df4')
```



DATASET4

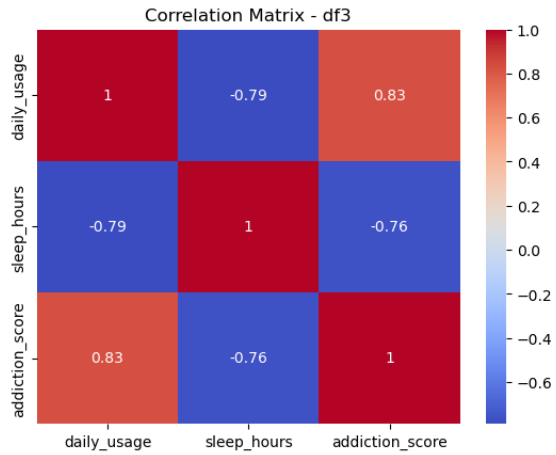
```
[169]: sns.histplot(df3_core['sleep_hours'], kde=True)
       plt.title("Sleep Distribution")
```

```
[169]: Text(0.5, 1.0, 'Sleep Distribution')
```



```
[170]: sns.heatmap(df3_core.corr(), annot=True, cmap='coolwarm')
       plt.title("Correlation Matrix - df3")
```

```
[170]: Text(0.5, 1.0, 'Correlation Matrix - df3')
```



STEP8:

- Normalization

Variable	Example Scale
daily_usage_hours	2–12 hours
phone_checks	20–200
sleep_deficit	0–4
addiction_score	1–10

--Phone checks (0–200) will dominate the formula
--Small-scale variables will barely affect the index
Therefore normalization is done 0≤x≤1

Min-Max Scaling
Formula:

$$X_{normalized} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

DF4

```
[171]: from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df4_scaled = df4_core.copy()

cols = ['daily_usage','phone_checks',
        'Screen_Time_Before_Bed','addiction_score']

df4_scaled[cols] = scaler.fit_transform(df4_scaled[cols])
```

```
[172]: df4_scaled.head()
```

[172]:

	daily_usage	phone_checks	Screen_Time_Before_Bed	addiction_score
0	0.347826	0.507692	0.538462	1.000000
1	0.478261	0.584615	0.346154	1.000000
2	0.504348	0.900000	0.192308	0.911111
3	0.269565	0.830769	0.538462	0.977778
4	0.217391	0.584615	0.384615	0.844444

DF3

```
[173]: df3_scaled = df3_core.copy()

cols = ['daily_usage','sleep_hours','addiction_score']

df3_scaled[cols] = scaler.fit_transform(df3_scaled[cols])
```

```
[175]: df3_scaled.head()
```

[175]:

	daily_usage	sleep_hours	addiction_score
0	0.528571	0.465517	0.857143
1	0.085714	0.637931	0.142857
2	0.642857	0.206897	1.000000
3	0.214286	0.551724	0.285714
4	0.428571	0.379310	0.714286

STEP9:

- Create BDI (Behavioural Dependency Index)

(Based on usage + phone checks + addiction)

Derive weights from correlation:

```
[176]: corr = df4_scaled.corr()['addiction_score']

usage_corr = abs(corr['daily_usage'])
checks_corr = abs(corr['phone_checks'])
bed_corr = abs(corr['Screen_Time_Before_Bed'])

total = usage_corr + checks_corr + bed_corr

w1 = usage_corr / total
w2 = checks_corr / total
w3 = bed_corr / total
```

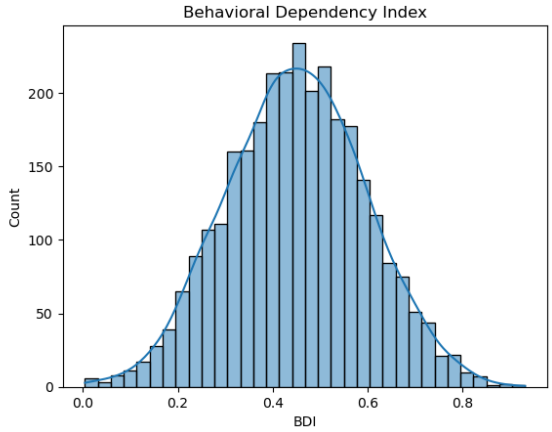
Construct BDI:

```
[177]: df4_scaled['BDI'] = (
           w1 * df4_scaled['daily_usage'] +
           w2 * df4_scaled['phone_checks'] +
           w3 * df4_scaled['Screen_Time_Before_Bed']
       )
```

Visualize BDI

```
[178]: sns.histplot(df4_scaled['BDI'], kde=True)
       plt.title("Behavioral Dependency Index")

[178]: Text(0.5, 1.0, 'Behavioral Dependency Index')
```



STEP10:

- Create RDI (Routine Disruption Index)

(Based on usage + sleep + academic impact)

Invert sleep:

```
[179]: df3_scaled['sleep_disruption'] = 1 - df3_scaled['sleep_hours']

[180]: df3_scaled.head()
```

[180]:

	daily_usage	sleep_hours	addiction_score	sleep_disruption
0	0.528571	0.465517	0.857143	0.534483
1	0.085714	0.637931	0.142857	0.362069
2	0.642857	0.206897	1.000000	0.793103
3	0.214286	0.551724	0.285714	0.448276
4	0.428571	0.379310	0.714286	0.620690

Derive weights:

```
[181]: corr = df3_scaled.corr()['addiction_score']

usage_corr = abs(corr['daily_usage'])
sleep_corr = abs(corr['sleep_disruption'])

total = usage_corr + sleep_corr

w1 = usage_corr / total
w2 = sleep_corr / total
```

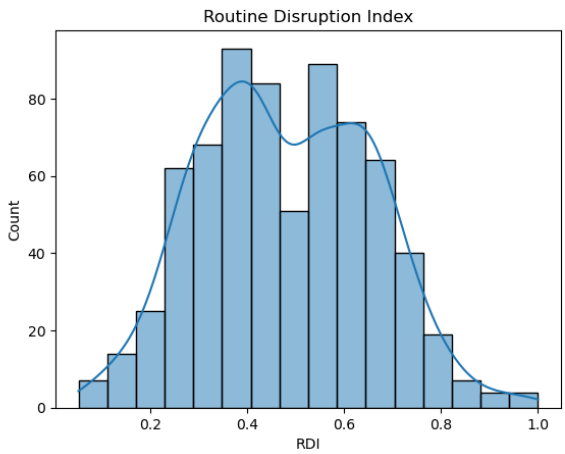
Construct RDI:

```
[182]: df3_scaled['RDI'] = (
    w1 * df3_scaled['daily_usage'] +
    w2 * df3_scaled['sleep_disruption']
)
```

Visualize RDI

```
[183]: sns.histplot(df3_scaled['RDI'], kde=True)
plt.title("Routine Disruption Index")

[183]: Text(0.5, 1.0, 'Routine Disruption Index')
```



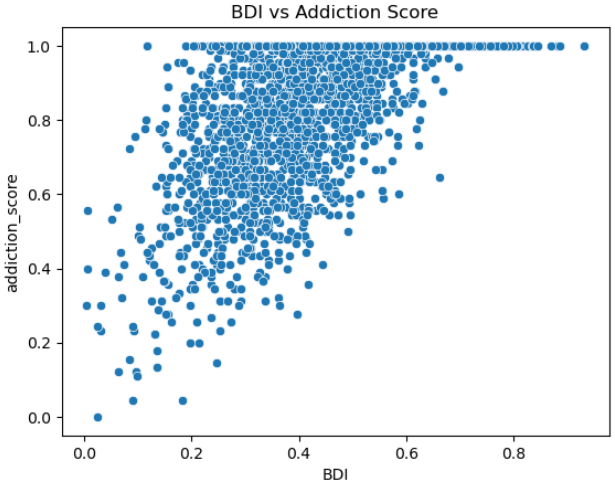
STEP11:

- Validation

BDI

```
[184]: sns.scatterplot(x='BDI', y='addiction_score', data=df4_scaled)
       plt.title("BDI vs Addiction Score")
```

```
[184]: Text(0.5, 1.0, 'BDI vs Addiction Score')
```



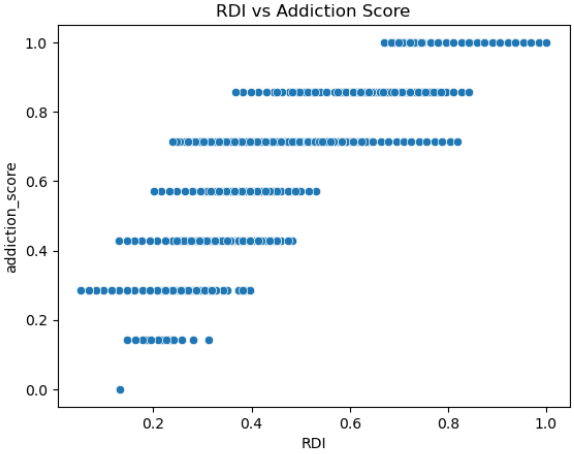

```
[185]: print(df4_scaled[['BDI','addiction_score']].corr())
```

```
                     BDI  addiction_score
BDI             1.000000         0.632044
addiction_score 0.632044         1.000000
```

RDI

```
[186]: sns.scatterplot(x='RDI', y='addiction_score', data=df3_scaled)
       plt.title("RDI vs Addiction Score")
```

```
[186]: Text(0.5, 1.0, 'RDI vs Addiction Score')
```




```
[187]: print(df3_scaled[['RDI','addiction_score']].corr())
```

```
                     RDI  addiction_score
RDI             1.000000         0.843929
addiction_score 0.843929         1.000000
```

STEP12:

- Categorize Indices

BDI

```
[188]: df4_scaled['BDI_Category'] = pd.cut(
           df4_scaled['BDI'],
           bins=[0,0.3,0.6,1],
           labels=['Low','Moderate','High']
       )
```

```
[189]: df4_scaled.head()
```

[189]:

	daily_usage	phone_checks	Screen_Time_Before_Bed	addiction_score	BDI	BDI_Category
0	0.347826	0.507692	0.538462	1.000000	0.396623	Moderate
1	0.478261	0.584615	0.346154	1.000000	0.506579	Moderate
2	0.504348	0.900000	0.192308	0.911111	0.612566	High
3	0.269565	0.830769	0.538462	0.977778	0.434457	Moderate
4	0.217391	0.584615	0.384615	0.844444	0.325148	Moderate

RDI

```
[190]: df3_scaled['RDI_Category'] = pd.cut(
           df3_scaled['RDI'],
           bins=[0,0.3,0.6,1],
           labels=['Low','Moderate','High']
       )
```

```
[191]: df3_scaled.head()
```

[191]:

	daily_usage	sleep_hours	addiction_score	sleep_disruption	RDI	RDI_Category
0	0.528571	0.465517	0.857143	0.534483	0.531403	Moderate
1	0.085714	0.637931	0.142857	0.362069	0.218082	Low
2	0.642857	0.206897	1.000000	0.793103	0.714822	High
3	0.214286	0.551724	0.285714	0.448276	0.326362	Moderate
4	0.428571	0.379310	0.714286	0.620690	0.520592	Moderate

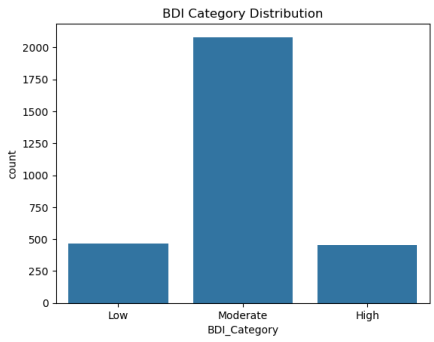
STEP13:

- Category Distribution

BDI

```
[192]: sns.countplot(x='BDI_Category', data=df4_scaled)
       plt.title("BDI Category Distribution")
```

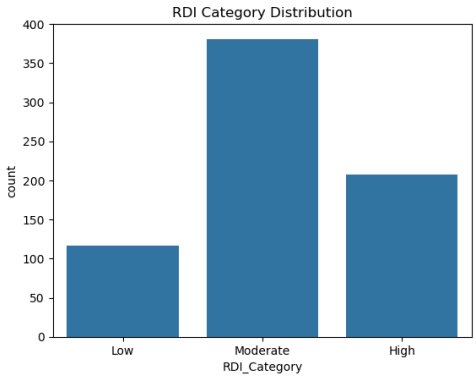
```
[192]: Text(0.5, 1.0, 'BDI Category Distribution')
```



RDI

```
[193]: sns.countplot(x='RDI_Category', data=df3_scaled)
       plt.title("RDI Category Distribution")
```

```
[193]: Text(0.5, 1.0, 'RDI Category Distribution')
```

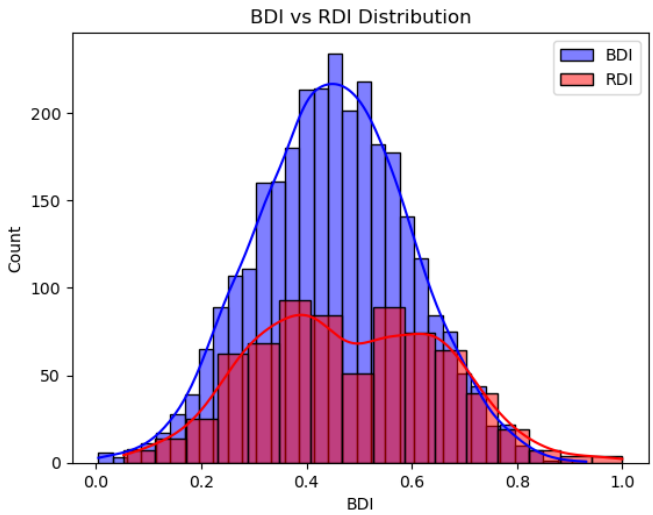


STEP14:

The Digital Usage Intensity Index (DUII) was constructed as a hierarchical composite metric integrating two primary dimensions: Behavioral Dependency Index (BDI) and Routine Disruption Index (RDI). The final DUII is defined as a weighted aggregation of these sub-indices, capturing both interaction intensity and lifestyle disruption effects of digital usage.

```
[194]: sns.histplot(df4_scaled['BDI'], color='blue', label='BDI', kde=True)
       sns.histplot(df3_scaled['RDI'], color='red', label='RDI', kde=True)
       plt.legend()
       plt.title("BDI vs RDI Distribution")
```

```
[194]: Text(0.5, 1.0, 'BDI vs RDI Distribution')
```



Final Conceptual Formula

$$DUII = \alpha\Big(w_1U + w_2C + w_3B\Big) + \beta\Big(w_4U + w_5S\Big)$$

- U = daily usages
- C = phone checks
- B = screen before bed
- S = sleep disruption

This shows layered construction.

- Finalizing the formula for the DUII

DUII =α(BDI)+ β(RDI)

Now finding α,β values

```
[195]: bdi_corr = df4_scaled[['BDI','addiction_score']].corr().iloc[0,1]
       rdi_corr = df3_scaled[['RDI','addiction_score']].corr().iloc[0,1]

       print("BDI correlation:", bdi_corr)
       print("RDI correlation:", rdi_corr)

       BDI correlation: 0.6320443091900633
       RDI correlation: 0.8439293203520548

[196]: total = abs(bdi_corr) + abs(rdi_corr)

       alpha = abs(bdi_corr) / total
       beta = abs(rdi_corr) / total

       print("Alpha (BDI weight):", alpha)
       print("Beta (RDI weight):", beta)

       Alpha (BDI weight): 0.4282219523028594
       Beta (RDI weight): 0.5717780476971405
```

Therefore

α= 0.43 β= 0.57

DUII =0.43(BDI)+ 0.57(RDI)

BDI=w1(daily_usage)+w2(phone_checks)+w3(screen_before_bed)

```
[201]: corr_series = df4_scaled[['daily_usage',
                                 'phone_checks',
                                 'Screen_Time_Before_Bed',
                                 'addiction_score']]\
                    .corr()['addiction_score']\
                    .drop('addiction_score')

       weights = abs(corr_series) / abs(corr_series).sum()

       print(weights)
```

```
daily_usage               0.697843
phone_checks              0.286146
Screen_Time_Before_Bed    0.016011
Name: addiction_score, dtype: float64
```

Now w1=0.70, w2=0.29, w3=0.01

RDI=w4(daily_usage)+w5(sleep_disruption)

Where => sleep_disruption=1−normalized(sleep_hours)

```
[203]: corr_series = df3_scaled[['daily_usage',
                                 'sleep_disruption',
                                 'addiction_score']] \
                    .corr()['addiction_score'] \
                    .drop('addiction_score')

       weights = abs(corr_series) / abs(corr_series).sum()

       print(weights)
```

```
daily_usage         0.521023
sleep_disruption    0.478977
Name: addiction_score, dtype: float64
```

Now w4=0.52 , w5=0.48

Final Formula

DUII= 0.43(0.7(daily_usage)+0.29(phone_checks)+0.01(screen_before_bed))+0.57(0.52 (daily_usage)+0.48(sleep_disruption))

DUII= 0.43 (0.7U + 0.29C + 0.01 B) + 0.57(0.52U + 0.48S)

U = normalized daily_usage

C = normalized phone_checks

B = normalized screen_before_bed

S = sleep_disruption

U→

```
[210]: usage_min = df4_core['daily_usage'].min()
       usage_max = df4_core['daily_usage'].max()
       print(usage_min)
       print(usage_max)

       0.0
       11.5
```

normalized(daily_usage)= (x - 0)/ (11.5 - 0)

- x= average daily_usage by the individual
- 0 = minimum daily_usage observed in dataset
- 11.5= maximum daily_usage observed in dataset

C→

```
[206]: checks_min = df4_core['phone_checks'].min()
       checks_max = df4_core['phone_checks'].max()
       print(checks_min)
       print(checks_max)

       20
       150
```

normalized(phone_checks)= (x - 20)/ (150 -20)

- x= average phone_checks by the individual
- 20 = minimum phone_checks observed in dataset
- 150= maximum phone_checks observed in dataset

B→

```
[207]: bed_min = df4_core['Screen_Time_Before_Bed'].min()
       bed_max = df4_core['Screen_Time_Before_Bed'].max()
       print(bed_min)
       print(bed_max)

       0.0
       2.6
```

normalized(screen_before_bed)= (x - 0)/ (2.6 - 0)

- x= average screen_before_bed by the individual
- 0 = minimum screen_before_bed observed in dataset
- 2.6 = maximum screen_before_bed observed in dataset

S→1−normalized(sleep_hours)

```
[209]: sleep_min = df3_core['sleep_hours'].min()
       sleep_max = df3_core['sleep_hours'].max()
       print(sleep_min)
       print(sleep_max)

       3.8
       9.6
```

normalized(sleep_hours)= (x – 3.8)/ (9.6 – 3.8)

- x= average hours slept by the individual
- 3.8 = minimum sleep observed in dataset
- 9.6 = maximum sleep observed in dataset

DUII (Digital Usage Intensity Index)

$$DUII = 0.43\underbrace{(0.7U + 0.29C + 0.01B)}_{\substack{BDI \\ \text{(Behavioural Dependency Index)}}} + 0.57\underbrace{(0.52U + 0.48S)}_{\substack{RDI \\ \text{(Routine Disruption Index)}}}$$

U = normalized daily usage

C = normalized phone_checks

B = normalized screen_before_bed

S = sleep disruption

now substitution the normalized equation of U, C, B, S in the DUII formula

$$\Rightarrow 0.43\left(0.7\left(\frac{usage}{11.5}\right) + 0.29\left(\frac{checks - 20}{130}\right) + 0.01\left(\frac{bed_hour_phone}{2.6}\right)\right)$$

$$+ 0.57\left(0.52\left(\frac{usage}{11.5}\right) + 0.48\left(1 - \frac{(sleep_hours - 3.8)}{5.8}\right)\right)$$

$\Rightarrow$ DUII Final Formula

DUII Range	Category	Meaning
0.00 – 0.30	Low	Healthy digital usage
0.30 – 0.60	Moderate	Controlled but high exposure
0.60 – 0.80	High	Risk of dependency
0.80 – 1.00	Severe	Strong digital intensity